

Software Requirements Specification

CanPlan 2.0: Cloud-Based Expansion

Draft SRS for Mobile App, Supporter Portal, and Serverless AWS Backend

Field	Value
Project	CanPlan 2.0
Prepared for	Capstone project planning and client review
Prepared by	Project Team
Date	May 28, 2026
Version	0.1 Draft
Technology Stack	React Native + TypeScript, React + TypeScript, Cognito, AppSync GraphQL, Lambda TypeScript, DynamoDB, S3, Amplify Hosting, CloudWatch, CDK or Amplify Gen 2

Document Status: Draft for discussion. Requirements should be validated with the client before implementation.

Revision History

Version	Date	Author	Description
0.1	May 28, 2026	Project Team	Initial SRS draft based on CanPlan 2.0 project brief, selected tech stack, and CanPlan 1.0 public app reference.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for CanPlan 2.0, a cloud-based expansion of the existing CanPlan task management app. The document describes the expected users, core features, user stories, functional requirements, non-functional requirements, system interfaces, data requirements, and acceptance criteria.

The SRS is intended for the project team, client stakeholders, designers, developers, testers, and future teams who may continue development after handoff.

1.2 Scope

CanPlan 2.0 will support primary users who need help completing tasks through clear step-by-step instructions. It will also support caregivers, teachers, employers, clinicians, and other support people who create, assign, monitor, and update tasks remotely.

- A cross-platform mobile app for iOS and Android.
- A web-based frontend portal for support people and administrators.
- Cloud-based user accounts, storage, synchronization, and role-based access.
- Offline access to assigned tasks and locally created content.
- Task creation using photos, captions, audio, and video files.
- Remote task assignment, progress tracking, and report generation.
- Accessibility support, including screen reader compatibility, large touch targets, and future Bluetooth switch input support.

1.3 Product Context

CanPlan 1.0 is a productivity app from the University of Victoria that helps users with cognitive challenges complete tasks by breaking activities into photo-based steps with optional text and audio. Its App Store listing describes key features such as task sequencing, scheduling, prompting, active task reminders, task categories, and support for iPhone and iPad. The current CanPlan 2.0 project brief states that CanPlan 1.0 uses local storage only, which limits synchronization, sharing, and support-person connectivity. CanPlan 2.0 will preserve the simple task-building experience while adding cloud accounts and remote support features.

1.4 References

ID	Source	Notes
R1	CanPlan 2.0 Requirements - With Phases PDF	Primary client project brief used for product vision, target users, key features, technical requirements, and roadmap.
R2	Apple App Store listing for CanPlan	Public CanPlan 1.0 reference used to understand existing app behavior, task sequencing, scheduling, audio/text/photo support, and app positioning.

2. Overall Description

2.1 Product Perspective

CanPlan 2.0 is a multi-client, cloud-backed system. The mobile app will be the main experience for primary users. The web portal will be the main experience for support people and administrators. AWS serverless services will provide authentication, API access, business logic, data storage, file storage, hosting, logging, monitoring, and infrastructure automation.

Layer	Selected Technology	Responsibility
Mobile App	React Native + TypeScript	Primary user task execution, task creation, offline cache, media capture, reminders, accessibility features.
Frontend Portal	React + TypeScript	Supporter dashboard, task builder, user/group management, reporting, admin workflows.
Authentication	Amazon Cognito	User sign-up, sign-in, password reset, roles, groups, and optional future SSO.
API	AWS AppSync GraphQL	Single API layer for mobile and portal clients; integrates with Lambda and DynamoDB resolvers.
Backend Logic	AWS Lambda with TypeScript	Business rules, task assignment, report generation, permission checks, notification workflows.
Database	Amazon DynamoDB	NoSQL storage for users, tasks, steps, schedules, groups, progress, and audit events.
File Storage	Amazon S3	Photos, audio, video, generated PDF reports, and other user-uploaded content.
Hosting	AWS Amplify Hosting	Build and host the web portal and related frontend environments.
Monitoring	Amazon CloudWatch	Logs, metrics, alarms, dashboards, and operational visibility.
IaC	AWS CDK or Amplify Gen 2	Repeatable infrastructure deployment and environment management.

2.2 User Classes

User Class	Description	Main Goals
Primary User	A person who uses CanPlan to complete tasks. This may include people with acquired brain injuries, dementia, cognitive barriers, behavior barriers, or diverse abilities.	View tasks, follow steps, receive reminders, ask for help, complete tasks independently.
Support Person	A caregiver, family member, teacher, employer, therapist, clinician, or job coach who supports one or more primary users.	Create tasks, assign tasks, review progress, provide just-in-time support.
Organization Admin	A user who manages an organization, school, employer program, or care team.	Manage users, groups, roles, templates, access, and reporting.
System Admin	A technical or operational administrator responsible for configuration and support.	Monitor system health, manage environments, view logs, and support releases.

2.3 Assumptions and Dependencies

- The project will start with the selected AWS serverless technology stack unless the client requests a different approach.
- The mobile app must work offline for previously synced tasks and media.
- The cloud platform must support at least 10,000 users as a target scalability requirement.
- The team will document security and privacy design, but formal penetration testing and Uvic privacy assessment are outside the student project scope unless otherwise agreed.
- Bluetooth switch accessibility and terminal mode may require platform-specific native modules or device management features and should be validated early.
- Generated PDF reports should be treated as sensitive files and protected by role-based access control.

3. User Stories

The following user stories describe the main product behavior from the perspective of each user class.

ID	Role	User Story	Priority
US-01	Primary User	As a primary user, I want to sign in securely so that I can access my tasks from supported devices.	Must
US-02	Primary User	As a primary user, I want to see today's tasks in a simple list so that I know what to do next.	Must

US-03	Primary User	As a primary user, I want to open a task and follow photo-based steps so that I can complete the task independently.	Must
US-04	Primary User	As a primary user, I want each step to support text, audio, and video so that instructions match my needs.	Must
US-05	Primary User	As a primary user, I want the app to work offline so that I can still follow tasks without internet access.	Must
US-06	Primary User	As a primary user, I want reminders and prompts so that I can start tasks at the right time.	Should
US-07	Primary User	As a primary user, I want to request help during a task so that a support person can guide me when I get stuck.	Should
US-08	Support Person	As a support person, I want to create a task with ordered steps so that I can prepare clear instructions for a user.	Must
US-09	Support Person	As a support person, I want to assign a task to one user or a group so that I can manage support at scale.	Must
US-10	Support Person	As a support person, I want to monitor task initiation and completion so that I can understand user progress.	Must
US-11	Support Person	As a support person, I want to edit a task remotely so that users receive updated guidance without needing my physical device.	Must
US-12	Support Person	As a support person, I want to send personal voice reminders so that reminders feel familiar and supportive.	Could
US-13	Support Person	As a support person, I want to export a PDF report so that I can share progress in clinical or educational review.	Should
US-14	Organization Admin	As an organization admin, I want to manage users and groups so that each support person only sees assigned users.	Must
US-15	Organization Admin	As an organization admin, I want setup modules such as Employment, Education, and Seniors/Dementia so that onboarding matches each context.	Should
US-16	System Admin	As a system admin, I want logs and metrics so that I can detect failures and monitor system health.	Must

4. Functional Requirements

4.1 Account and Access Management

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-AUTH-01	The system shall allow users to create an account using email and password.	Must	M1	A new user can register, verify the account, and sign in.
FR-AUTH-02	The system shall support secure sign-in, sign-out, and password reset through Amazon Cognito.	Must	M1	A user can reset a forgotten password and regain access.
FR-AUTH-03	The system shall support role-based access for primary users, support people, organization admins, and system admins.	Must	M1	Each role can access only the screens and data allowed for that role.
FR-AUTH-04	The system shall support linking a support person account to a primary user account after consent or approval.	Must	M2	A support person can only view and manage a linked user after a valid link exists.
FR-AUTH-05	The system should allow future SSO integration if the client requires school, employer, or organization login.	Should	M3	Architecture notes identify Cognito federation or hosted UI as the expected path.

4.2 Task and Step Management

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-TASK-01	The system shall allow users or support people to create a task with a title, category, description, schedule, and ordered steps.	Must	M1	A task can be created and retrieved from both mobile and portal clients.
FR-TASK-02	The system shall allow each step to include text instructions.	Must	M1	A step can show readable text instructions on mobile.
FR-TASK-03	The system shall allow each step to include photos stored in S3.	Must	M1	A step can display an uploaded image from secure storage.
FR-TASK-04	The system should allow each step to include audio instructions stored in S3.	Should	M2	A step can play uploaded audio on mobile.
FR-TASK-05	The system should allow each step to include video instructions stored in S3.	Should	M2	A step can play uploaded video on mobile.
FR-TASK-06	The system shall allow task steps to be reordered.	Must	M1	A user can drag or otherwise reorder steps and save the new order.
FR-TASK-07	The system shall allow tasks to be assigned to a primary user.	Must	M2	Assigned tasks appear in the primary user mobile app after sync.
FR-TASK-08	The system shall allow tasks to be assigned to a group of users.	Should	M2	A support person can assign one task set to multiple users.

FR-TASK-09	The system shall allow task categories and templates.	Should	M2	A support person can reuse a task template across users.
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4.3 Scheduling, Reminders, and Task Execution

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-SCHED-01	The system shall show today's scheduled tasks in the mobile app.	Must	M1	A primary user can see tasks due today in a simple view.
FR-SCHED-02	The system shall allow one-time and recurring task schedules.	Should	M2	Daily, weekly, and custom recurrence rules can be configured.
FR-SCHED-03	The system shall support reminders for scheduled tasks.	Should	M2	The mobile app can alert the user when a task should start.
FR-SCHED-04	The system shall allow users to mark a task as started, paused, resumed, skipped, or completed.	Must	M2	Progress events are saved locally and synced to the cloud.
FR-SCHED-05	The system should support active task reminders while a task is in progress.	Should	M3	The user receives a prompt if the task remains active longer than expected.

4.4 Offline Mode and Sync

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-OFF-01	The mobile app shall cache assigned tasks, schedules, steps, and required media for offline use.	Must	M2	A user can open previously synced tasks while offline.
FR-OFF-02	The mobile app shall record task progress while offline.	Must	M2	Task progress events created offline sync after connectivity returns.
FR-OFF-03	The system shall define conflict handling for task edits and progress events.	Must	M2	A documented rule handles conflicting updates, such as last-write-wins for metadata and append-only progress logs.
FR-OFF-04	The mobile app should show clear sync status to the user or support person.	Should	M2	The app indicates when content is synced, pending, or failed.

4.5 Supporter Portal

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-PORTAL-01	The portal shall allow support people to view assigned users.	Must	M1	Support people only see users they are authorized to support.
FR-PORTAL-02	The portal shall provide a task builder for creating and editing tasks and steps.	Must	M1	A support person can create a task with ordered media-rich steps.
FR-PORTAL-03	The portal shall support pushing tasks to individual users.	Must	M2	A task assigned in the portal appears on the correct mobile account.
FR-PORTAL-04	The portal should support group task assignment.	Should	M2	A support person can assign a task to a configured group.
FR-PORTAL-05	The portal shall display high-level progress data.	Must	M2	The dashboard shows task status, starts, completions, and missed tasks.
FR-PORTAL-06	The portal should generate PDF reports from progress data.	Should	M3	A report can be generated, stored securely, and downloaded by authorized users.

4.6 Real-Time Support and Communication

ID	Requirement	Priority	Phase	Acceptance Criteria
FR-HELP-01	The mobile app should allow a primary user to request help during a task.	Should	M3	A support person receives a visible help request.
FR-HELP-02	The system should support text-based in-app messaging between a primary user and linked support person.	Should	M3	Messages are delivered to the correct linked accounts and logged securely.
FR-HELP-03	The system could support personal voice reminders from a family member or caregiver.	Could	M3	A support person can record and assign a reminder audio file.

4.7 Accessibility and Device Modes

ID	Requirement	Priority	Phase	Acceptance Criteria
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FR-ACC-01	The mobile app and portal shall follow WCAG-aligned accessibility practices.	Must	M1-M3	Screens pass accessibility review for labels, contrast, keyboard access, and screen reader support where applicable.
FR-ACC-02	The mobile app shall support portrait and landscape orientation.	Must	M2	Core screens render correctly in both orientations.
FR-ACC-03	The mobile app should support Bluetooth accessibility switch input.	Should	M3	A switch user can navigate core task execution flows.
FR-ACC-04	The mobile app should support terminal mode or a guided-access-style restricted mode.	Should	M3	A configured device can limit user access to CanPlan during use.
FR-ACC-05	The system should include setup modules for Employment, Education, and Seniors/Dementia.	Should	M3	The setup wizard recommends module-specific defaults and task templates.

5. Non-Functional Requirements

ID	Requirement	Priority	Phase	Acceptance Criteria
NFR-PERF-01	The system shall remain responsive for the target scale of 10,000+ users.	Must	M2-M3	Load tests show acceptable latency for core API flows under agreed test load.
NFR-PERF-02	Core mobile screens should load cached content within 2 seconds on supported devices.	Should	M2	Offline task list and task detail screens meet the target on test devices.
NFR-SEC-01	The system shall encrypt data in transit using HTTPS/TLS.	Must	M1	All client API and file access uses secure transport.
NFR-SEC-02	The system shall encrypt sensitive data and files at rest using AWS-managed encryption, with AES-256 where supported.	Must	M2	DynamoDB and S3 encryption are enabled and documented.
NFR-SEC-03	The system shall enforce least-privilege access for users and AWS resources.	Must	M2	IAM roles, Cognito groups, and API authorization rules restrict access.
NFR-PRIV-01	The system shall be designed with FIPPA compliance considerations.	Must	M2	Data residency, access control, audit logging, and retention decisions are documented.
NFR-REL-01	The system shall log operational errors and key backend events.	Must	M2	CloudWatch logs exist for Lambda and AppSync integration errors.
NFR-OBS-01	The system shall provide CloudWatch dashboards or metrics for API errors, Lambda errors, latency, and storage usage.	Must	M3	A project dashboard shows health indicators for production-like environments.
NFR-MAINT-01	Infrastructure shall be defined using AWS CDK or Amplify Gen 2.	Must	M1-M3	A new environment can be deployed from documented commands.
NFR-COMP-01	The mobile app shall support modern iOS and Android versions selected by the team and approved by the client.	Must	M1	Supported OS versions are documented and tested.
NFR-USAB-01	The mobile UI shall minimize visual complexity and avoid unnecessary distractions.	Must	M1-M3	Task execution screens show only the information needed for the current step.

6. Architecture and Interfaces

6.1 Logical Architecture

The system is expected to use the following high-level request flow:

- React Native mobile app and React portal authenticate users through Amazon Cognito.
- Authenticated clients call AWS AppSync GraphQL queries and mutations.
- AppSync resolves simple data operations directly through DynamoDB where appropriate.
- AppSync calls TypeScript Lambda functions for business rules, permission checks, report generation, media processing, and complex workflows.
- DynamoDB stores structured application data.
- S3 stores photos, audio, video, and generated reports.
- CloudWatch stores logs, metrics, dashboards, and alarms.

6.2 External Interface Requirements

Interface	Requirement
Mobile UI	Must support touch-first task execution, large targets, screen reader labels, simple navigation, portrait and landscape layouts, and offline status.

Portal UI	Must support keyboard navigation, accessible forms, dashboards, task builder workflows, user management, and report access.
GraphQL API	Must expose typed queries, mutations, and subscriptions where needed for tasks, users, assignments, progress, media metadata, and reports.
Authentication	Must use Cognito access tokens to authorize API calls and protect private resources.
File Uploads	Must use secure upload patterns for S3, such as pre-signed URLs or Amplify storage integration with authorization rules.
Monitoring	Must send logs and metrics to CloudWatch for debugging, audits, and health monitoring.

6.3 Preliminary GraphQL Domain Model

Entity	Key Fields	Purpose
UserProfile	userId, role, displayName, organizationId, accessibilitySettings	Stores user profile and role metadata.
SupportLink	supporterId, primaryUserId, status, permissions	Represents approved supporter-user relationships.
Task	taskId, ownerId, title, categoryId, description, scheduleRule, status	Represents a reusable or assigned task.
TaskStep	stepId, taskId, order, text, mediaRefs, expectedDuration	Represents one ordered instruction step.
Assignment	assignmentId, taskId, userId/groupId, dueDate, recurrence, active	Tracks which users receive which tasks.
ProgressEvent	eventId, assignmentId, userId, eventType, timestamp, source	Append-only telemetry for starts, pauses, skips, completions, and sync state.
MediaAsset	assetId, s3Key, type, ownerId, size, createdAt	Metadata for uploaded images, audio, and video.
Report	reportId, scope, dateRange, s3Key, createdBy, createdAt	Generated PDF reports.

7. Data, Privacy, and Security Requirements

7.1 Data Classification

Data Type	Examples	Sensitivity	Expected Storage
Account Data	Email, display name, role, organization link	Sensitive	Cognito and DynamoDB
Task Data	Task titles, categories, instructions, schedules	Sensitive	DynamoDB
Media Data	Photos, audio, video instructions, personal reminders	Sensitive	S3
Progress Data	Task started, completed, skipped, timestamps	Sensitive	DynamoDB
Reports	PDF summaries for clinical or educational review	Highly sensitive	S3 with restricted access
Operational Logs	Errors, request metadata, service metrics	Internal	CloudWatch

7.2 Security Controls

- Use Cognito authentication for all non-public app and portal access.
- Use AppSync authorization rules to enforce role-based and owner-based access.
- Use least-privilege IAM roles for Lambda, AppSync, DynamoDB, S3, and CloudWatch.
- Encrypt DynamoDB tables and S3 buckets at rest.
- Use HTTPS/TLS for all client communication.
- Avoid storing private user data in client logs, analytics events, or error messages.
- Record access and modification events needed for auditability without over-collecting sensitive data.
- Define data retention and deletion requirements with the client before production release.

8. Acceptance Criteria and Testing Strategy

8.1 Acceptance Criteria by Module

Module	Acceptance Criteria
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Module 1: Technical Framework Specification and UI/UX Design	Architecture is documented; initial GraphQL schema is drafted; UI/UX flows exist for mobile and portal; accessibility plan is documented; proof-of-concept screens run locally or in a development environment.
Module 2: Core Development	Cognito auth works; AppSync API is deployed; DynamoDB and S3 are configured; core task creation, assignment, offline cache, and progress sync are functional; security controls are documented.
Module 3: Integration and Production Release	Mobile and portal are integrated with backend; QA test cases pass; accessibility checks pass; CloudWatch dashboards exist; PDF reporting and operational handoff documentation are prepared.

8.2 Testing Types

- Unit tests for TypeScript utility functions, Lambda handlers, and data validation.
- Component tests for React Native and React UI components.
- Integration tests for AppSync, Lambda, DynamoDB, S3, and Cognito flows.
- Offline sync tests for network loss, delayed upload, and conflict handling.
- Accessibility tests using screen reader checks, keyboard navigation checks, contrast checks, and large text checks.
- Performance and load tests for core API flows against the 10,000+ user scalability target.
- Security review for authorization rules, file access, token handling, and log redaction.
- User acceptance testing with representative user groups and support people, subject to client coordination.

9. Requirements Traceability Matrix

Feature Area	User Stories	Functional Requirements	Non-Functional Requirements
Account creation and cloud access	US-01	FR-AUTH-01, FR-AUTH-02, FR-AUTH-03	NFR-SEC-01, NFR-SEC-03
Task sequencing with media	US-02, US-03, US-04, US-08	FR-TASK-01 to FR-TASK-06	NFR-USAB-01, NFR-COMP-01
Offline mode	US-05	FR-OFF-01 to FR-OFF-04	NFR-PERF-02, NFR-REL-01
Remote support and task assignment	US-09, US-10, US-11	FR-TASK-07, FR-TASK-08, FR-PORTAL-01 to FR-PORTAL-05	NFR-SEC-03, NFR-OBS-01
Reporting	US-13	FR-PORTAL-06	NFR-PRIV-01, NFR-SEC-02
Accessibility	US-15	FR-ACC-01 to FR-ACC-05	NFR-USAB-01, NFR-COMP-01
Monitoring and operations	US-16	FR-PORTAL-05	NFR-REL-01, NFR-OBS-01, NFR-MAINT-01

10. Risks and Open Questions

ID	Risk / Question	Impact	Recommended Action
RQ-01	Should CanPlan 2.0 include AI or agent-based task assistance, or remain manually configured?	Affects scope, privacy, safety, cost, and data design.	Ask client to confirm before designing AI features.
RQ-02	Will the team build on CanPlan 1.0 code or start fresh?	Affects migration, design reuse, and technical constraints.	Request access to existing repository or documentation if reuse is expected.
RQ-03	Which AWS account should host development, staging, and production?	Affects deployment ownership, billing, security, and handoff.	Confirm whether the client provides AWS accounts or the team creates temporary environments.
RQ-04	What data residency and FIPPA requirements apply to production data?	Affects AWS region, storage, logging, and backups.	Confirm preferred AWS region and privacy review process.
RQ-05	What is the minimum supported iOS and Android version?	Affects React Native dependencies and accessibility testing.	Agree on support matrix during Module 1.
RQ-06	What level of terminal mode is required?	May require OS-level mobile device management rather than only app code.	Validate feasibility for iOS and Android early.
RQ-07	What Bluetooth switch devices must be supported?	Device compatibility may require native modules and testing hardware.	Ask client for target devices and accessibility scenarios.
RQ-08	What counts as a complete final delivery?	Affects acceptance criteria and project planning.	Confirm whether final handoff requires production deployment, demo, documentation, or all of these.

11. Glossary

Term	Meaning
Primary User	The person using the mobile app to complete tasks.
Support Person	A caregiver, teacher, employer, clinician, family member, or other helper who supports a primary user.
Task	A sequence of steps that helps a user complete an activity.
Step	One instruction inside a task. A step may include text, photo, audio, or video.
Telemetry	High-level usage and progress data such as task started, completed, skipped, or delayed.
Offline Mode	The ability to use cached tasks and record progress without an internet connection.
Terminal Mode	A restricted app/device mode that limits distractions and keeps the user focused on CanPlan.
WCAG	Web Content Accessibility Guidelines, a standard for accessible digital experiences.